ELECTRICAL FAN CONTROL [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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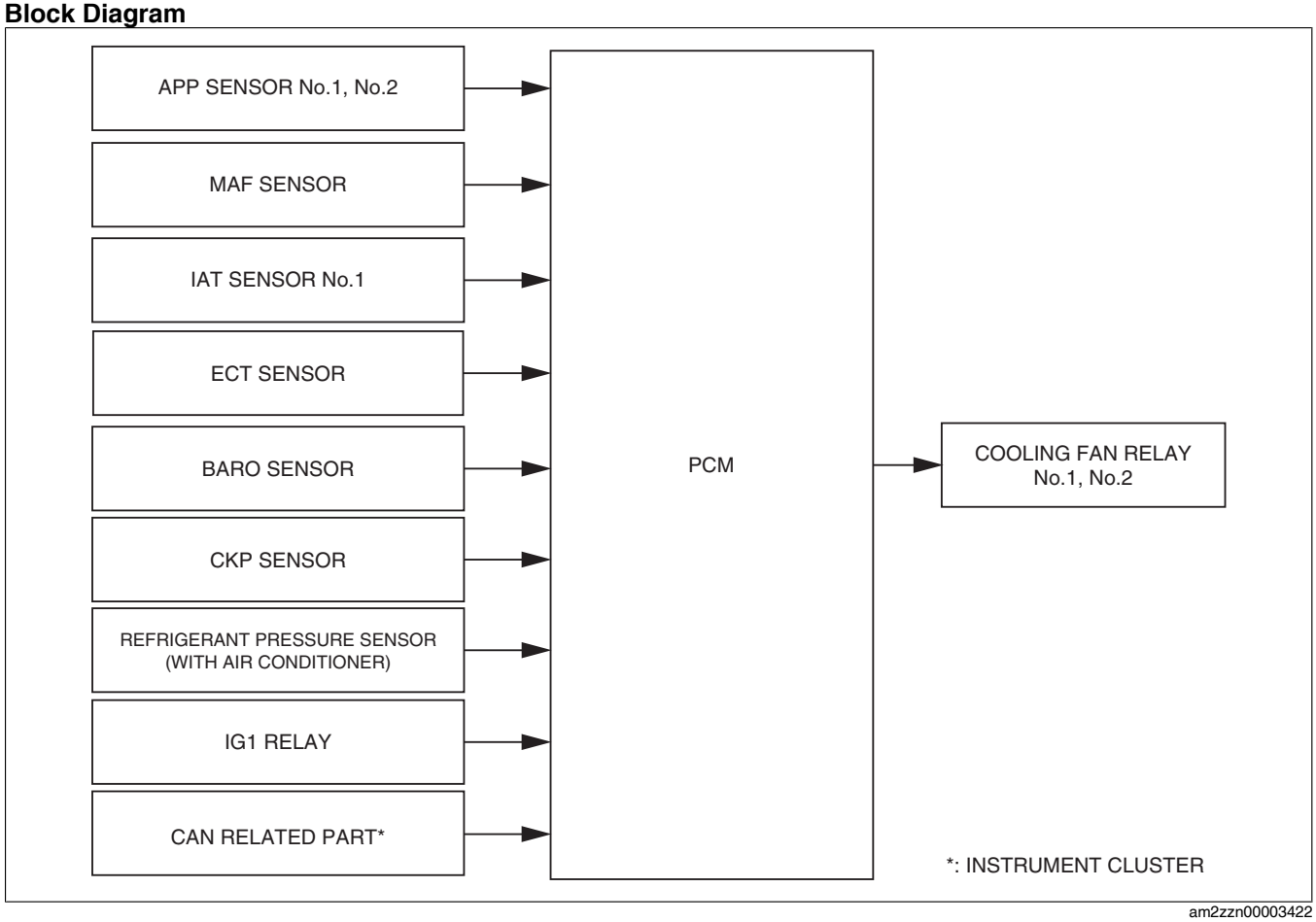
Outline

- Through cooling of the radiator and condenser by operation of the cooling fan (according to vehicle conditions), engine reliability and cooling performance have been improved.
- The PCM determines the engine operation conditions (based on the signals from each sensor), and turns cooling fan relays No.1 and No.2 on/off to control the rotation speed of the cooling fan motor.

Block Diagram



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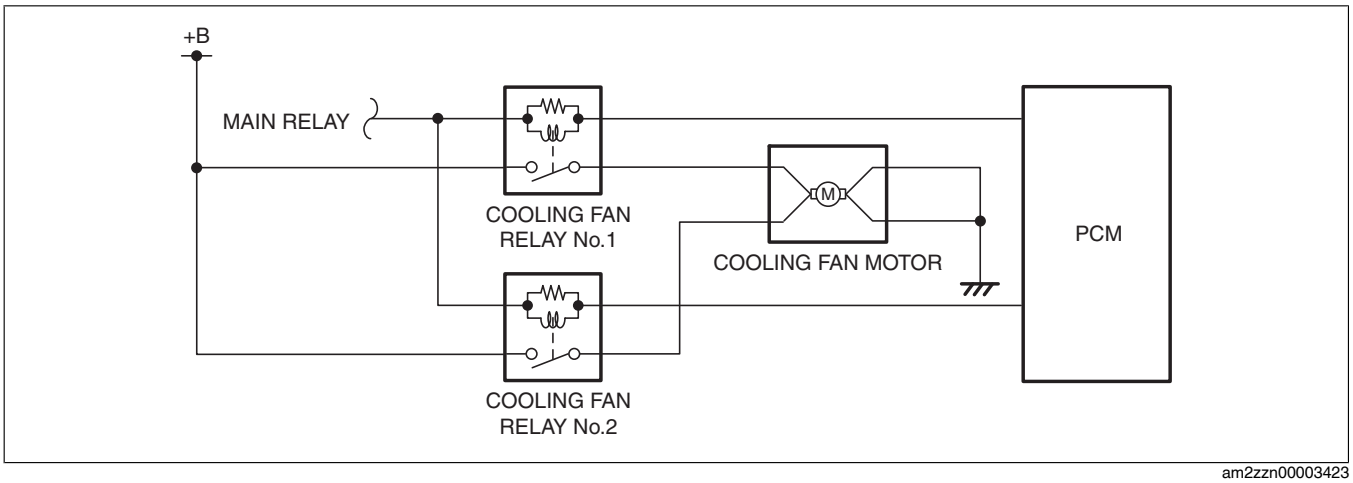
Operation

- The PCM controls the cooling fan motor rotation speed by turning cooling fan relays No.1 and No.2 on/off based on each demand airflow volume.

Operation Pattern

Demand airflow volume	Cooling fan relay		Cooling fan motor
	No.1	No.2	
No airflow	OFF	OFF	Stop
Small	ON	OFF	Low-speed
Large	ON	ON	High-speed



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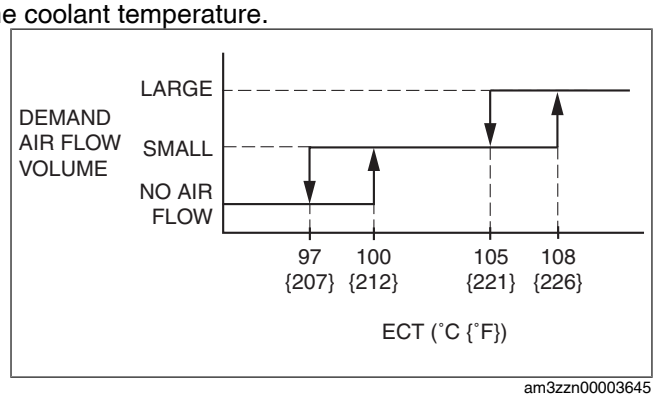
Engine coolant temperature condition

- Changes the demand airflow volume according to the engine coolant temperature.
- To suppress overheating, the cooling fan is operated when the engine coolant temperature is high. The demand airflow volume is large or small depending on the engine coolant temperature.



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A/C operation condition (with air conditioner)

- Controls the demand airflow volume depending on the refrigerant pressure and vehicle speed when the A/C switch is on.

Operation condition	Demand airflow volume
• When any of the following conditions is met. – Refrigerant pressure: **1.3 MPa {13 kgf/cm^2, 189 psi} or less**, vehicle speed: **45 km/h {28 mph} or more** – Refrigerant pressure: **2.2 MPa {22 kgf/cm^2, 319 psi} or less**, vehicle speed: **85 km/h {53 mph} or more**	No airflow
• When any of the following conditions is met. – Refrigerant pressure: **1.3 MPa {13 kgf/cm^2, 189 psi} or less**, vehicle speed: **45 km/h {28 mph} or less** – Refrigerant pressure: **1.3—1.5 MPa {14—15 kgf/cm^2, 189—217 psi}**, vehicle speed: **85 km/h {53 mph} or less** – Refrigerant pressure: **1.5—2.2 MPa {16—22 kgf/cm^2, 218—319 psi}**, vehicle speed: **65—85 km/h {41—52 mph}**	Small
• When any of the following conditions is met. – Refrigerant pressure: **1.5—2.2 MPa {16—22 kgf/cm^2, 218—319 psi}**, vehicle speed: **65 km/h {40 mph} or less** – Refrigerant pressure: **2.2 MPa {22 kgf/cm^2, 319 psi} or more**	Large

After Cooling Control

- After continuous, high-load driving, the cooling fan operates (demand airflow volume: small) for up to **9 min** under the following conditions to suppress temperature increase inside the engine compartment.
  - Driving on ascending roads or after continuing to drive under high-load conditions
  - After continuous, sudden acceleration from standstill and repeated, sudden acceleration
  - Engine coolant temperature: **80 °C {176 °F} or more**

Conditions during collision

- If a collision signal from the SAS control module is received, the PCM stops the engine cooling fan to avoid use of battery voltage.